# Notes: Plantin (RFS, 2015)

**Shadow Banking and Bank Capital Regulation**

## Contents

# 1 Big picture

- **Question.** How should a bank regulator set capital requirements when banks can partly avoid these requirements through opaque shadow-banking transactions?
- **Motivation.** Traditional banks issue deposits and other money-like liabilities. These claims are useful because households and firms use them to settle trades. But if these claims become risky, the risk does not only hurt the bank's own customers; it also disrupts real transactions between the bank's customers and other agents. Banks do not internalize this full social cost.
- **Baseline idea.** Without regulation, banks choose excessive leverage because they value cash today and do not fully internalize the real costs created when their money-like liabilities become risky. A benevolent regulator therefore imposes a binding risk-adjusted leverage cap, or equivalently a capital requirement.
- **Central twist.** Capital requirements may be imperfectly enforceable. Banks may pledge additional claims on their assets through a shadow-banking sector that the regulator cannot observe. In the paper, this sector is represented by money market funds (MMFs) that issue near-money claims to households and buy bank-risk exposures outside the regulatory perimeter.
- **Formal banking versus shadow banking.** Formal banking allows ex ante commitment: before private information is realized, the bank can commit to a common financing policy that pools future bank types. Shadow banking is a spot market: once the bank has private information about asset quality, it trades with the MMF. This creates adverse selection in the shadow sector.
- **Key mechanism.** Tightening formal capital requirements lowers regulated leverage, but it also gives the bank stronger incentives to raise cash through shadow banking. Since the shadow market is subject to a lemons discount, the bank must transfer more asset risk per dollar raised in the shadow sector than it would in the regulated sector.
- **Main result.** Stricter capital requirements can be counterproductive. Over a relevant range, a tighter requirement is Pareto dominated by a looser one: the bank ends up with the same cash, but households bear more risk because the bank substitutes from regulated funding to shadow funding.
- **Policy implication.** The regulator faces two possible local responses to regulatory arbitrage. One response is to tighten capital requirements and accept a larger shadow-banking sector. The other is to relax requirements enough to bring the activity back inside the regulated banking sector. The second response can be optimal.
- **Role of adverse selection.** Adverse selection in shadow banking has two opposite effects. It makes shadow funding costly, which curbs shadow activity. But it also makes each dollar of shadow funding require a larger transfer of risk. When regulation is chosen optimally, the first effect dominates in equilibrium: more adverse selection can improve welfare by limiting shadow banking. When regulation is inefficiently tight, adverse selection is costly because it makes regulatory arbitrage inefficient.
- **Takeaway.** Bank capital regulation cannot be evaluated only by looking at the formal banking sector. If activity can migrate to lightly regulated money-like intermediaries, tighter bank capital rules may increase total risk borne by holders of money-like claims.

# 2 Model environment and key objects

## 2.1 Agents and dates

The model is a two-date economy, with a preliminary regulatory date in the regulated versions. The core agents are:

- a **household**, endowed with the numeraire good and needing a store of value;
- an **entrepreneur**, who produces a second good valued only by the household;
- a **bank shareholder**, who can originate a risky loan portfolio and also has a nonpledgeable outside investment opportunity;
- an **MMF manager**, who can issue near-money claims and invest resources either in storage or in claims on the bank;
- a **regulator**, in the regulated versions, who grants a banking license and chooses a capital requirement.

There are two economically important dates:

- At date 1, the household stores resources with the bank and/or the MMF; the bank issues claims backed by its loan portfolio; and the entrepreneur chooses an initial production scale.
- At date 2, financial payoffs are realized; the entrepreneur observes the household's date-2 resources, adjusts capacity, and sells output to the household.

In the regulated versions, there is also a date 0 at which the regulator sets the permitted design of the bank's liabilities.

## 2.2 The real-side friction

The entrepreneur chooses an initial scale $N_1$ at date 1. At date 2, he can revise it to $N_2$, but changing scale is costly:

$$\frac{k}{2}(N_2 - N_1)^2, \qquad k > 0.$$

Each unit of output costs the entrepreneur a disutility $c \in (0, 1)$, while the household values each unit at one unit of the numeraire. Because $c < 1$, there are gains from trade. But the entrepreneur cannot commit at date 1 to produce at date 2. Ex post, he has all the bargaining power and charges the household the maximum price it is willing to pay.

The important implication is that uncertainty about the household's date-2 resources creates real costs. If the household's store of value is risky, the entrepreneur cannot choose date-1 capacity perfectly. The economy therefore cares about the riskiness of bank liabilities, not only their expected payoff.

## 2.3 The bank's loan portfolio

The bank can originate a loan portfolio that requires investment $I > 0$ at date 1. The portfolio payoff at date 2 is:

$$\begin{cases} L + l, & \text{if the portfolio is good, with probability } p, \\ L, & \text{if the portfolio is bad, with probability } 1 - p, \end{cases} \qquad 0 < L < L + l.$$

Thus $L$ is the safe part of the payoff and $l$ is the risky upside. The expected risky upside is $pl$.

The bank shareholder also has an outside investment opportunity. If he invests $x$ units at date 1, it produces:

$$\begin{cases} x + f(x), & \text{with probability } q, \\ x, & \text{with probability } 1 - q. \end{cases}$$

The return $f(x)$ is nonpledgeable. This assumption makes date-1 cash valuable to the bank shareholder: cash extracted from bank financing can be invested in a socially valuable but nonpledgeable opportunity.

## 2.4 Bank liabilities: the two-parameter security

A generic bank claim sold to the household is described by two numbers $\mu, \lambda \in [0, 1]$:

$$\text{payoff to household} = \begin{cases} \mu L, & \text{if the portfolio is bad,} \\ \mu L + \lambda l, & \text{if the portfolio is good.} \end{cases}$$

The household pays the expected value of this security at date 1:

$$\mu L + \lambda pl.$$

The interpretation is:

- $\mu$ measures how much of the safe cash flow $L$ is pledged.
- $\lambda$ measures how much of the risky cash flow $l$ is pledged.
- $\lambda$ is the key risk-adjusted leverage variable: a larger $\lambda$ transfers more portfolio risk to money-like liability holders.

Funding the loan requires:

$$\mu L + \lambda pl \geq I.$$

The excess cash $\mu L + \lambda pl - I$ is paid as a dividend to the bank shareholder and can be invested in the outside opportunity.

## 2.5 Formal banking and shadow banking

The model distinguishes two ways of transferring risk:

- **Formal banking**: the regulator observes the claim sold directly by the bank to the household and can impose an ex ante commitment. The bank can be required to pool across future private-information types.
- **Shadow banking**: the regulator cannot observe the bank's trade with the MMF. The MMF raises money-like liabilities from the household and buys claims on bank assets in a spot market after the bank has private information.

Thus, the core difference is not that shadow banking performs a different function. It performs the same money-creation function, but outside the regulatory perimeter and without the same commitment technology.

# 3 Models and theoretical structure (all equations + interpretation)

## 3.1 Date-2 trade and the real cost of risky money

Suppose the household has date-2 resources $W_2$ and the entrepreneur chose date-1 capacity $N_1$. The entrepreneur sets the output price equal to the household's willingness to pay, one per unit, and adjusts production to match the household's resources:

$$N_2 = W_2.$$

The entrepreneur's realized profit is:

$$(1-c)W_2 - \frac{k}{2}(W_2 - N_1)^2.$$

Because the entrepreneur chooses $N_1$ before $W_2$ is realized, he chooses:

$$N_1 = \mathbb{E}[W_2].$$

His expected utility is therefore:

$$U_E = (1-c)\mathbb{E}[W_2] - \frac{k}{2}\mathrm{Var}(W_2). \tag{1}$$

**Economic interpretation.** The first term is the expected surplus from trade. The second term is the real resource cost created by uncertainty in the household's purchasing power. Money-like liabilities are valuable because they help the household trade with the entrepreneur, but their riskiness reduces the efficiency of this trade.

## 3.2 Household wealth under a bank security

If the bank sells the security $(\mu, \lambda)$ to the household, only the risky part $\lambda l$ changes the variance of the household's date-2 resources. The household's date-2 resources are:

$$W_2 = \begin{cases} W + (1-p)\lambda l, & \text{with probability } p, \\ W - p\lambda l, & \text{with probability } 1-p. \end{cases}$$

Thus,

$$\mathbb{E}[W_2] = W, \qquad \mathrm{Var}(W_2) = p(1-p)\lambda^2 l^2.$$

Using equation (1), the entrepreneur's expected utility is:

$$U_E = W(1-c) - \frac{k}{2} p(1-p)\lambda^2 l^2. \tag{2}$$

**Economic interpretation.** The safe tranche $\mu L$ does not create real-side risk. The risky tranche $\lambda l$ does. Therefore, the externality from bank leverage is entirely captured by $\lambda$.

## 3.3 The bank shareholder's payoff

For a claim $(\mu, \lambda)$, the bank shareholder's expected utility is:

$$\begin{aligned} U_B &= \mu L + \lambda p l + (1-\mu)L + p(1-\lambda)l - I + qf(\mu L + \lambda p l - I) \\ &= L + pl - I + qf(\mu L + \lambda p l - I). \end{aligned} \tag{3}$$

The expression has two parts. The term $L + pl - I$ is the expected net payoff from the loan portfolio. The term $qf(\mu L + \lambda p l - I)$ is the expected gain from using date-1 cash to fund the outside opportunity.

Since $f$ is increasing, $U_B$ rises with both $\mu$ and $\lambda$. The unregulated bank therefore chooses:

$$\mu = 1, \qquad \lambda = 1. \tag{4}$$

**Economic interpretation.** The unregulated bank pledges the entire portfolio. It chooses maximum leverage because it values date-1 cash but does not internalize the full real-side cost created by risky money-like liabilities. This is the paper's microfoundation for excessive bank leverage.

## 3.4 Prudential regulation with perfect enforcement

A regulator who can perfectly enforce the bank's security design maximizes the utilitarian welfare of the nonbank agents plus the value created by the bank's outside investment. The welfare expression is:

$$L + pl - I + qf(\mu L + \lambda p l - I) + W(1-c) - \frac{k}{2} p(1-p)\lambda^2 l^2. \tag{5}$$

The regulator's trade-off is clear:

- a higher $\lambda$ raises more date-1 cash and increases the value of the bank shareholder's outside investment;
- a higher $\lambda$ also makes money-like liabilities riskier and raises real adjustment costs for the entrepreneur.

The optimal security is:

$$\mu^* = 1,$$

and the optimal risky pledge share $\lambda^* \in (0, 1)$ solves:

$$qf'(\lambda^* pl + L - I) = \lambda^* k(1 - p)l. \tag{6}$$

**Economic interpretation of equation (6).** The left-hand side is the marginal value of relaxing the capital requirement and giving the bank more cash to invest outside the bank. The right-hand side is the marginal real-side externality from making bank liabilities riskier. The regulator chooses $\lambda$ so that these are equal.

The regulation can be implemented as a capital requirement. The fraction of the portfolio payoff that can back household debt cannot exceed:

$$\frac{L + \lambda^* pl}{L + pl}. \tag{7}$$

This is a cap on risk-adjusted leverage. The safe part $L$ can be fully pledged, but only a fraction $\lambda^*$ of the risky payoff can be pledged.

## 3.5 Comparative statics of the perfect-enforcement capital requirement

Equation (6) gives the core comparative statics:

- If $q$ rises, the outside opportunity is more likely to be productive, so bank cash is more valuable and the optimal $\lambda^*$ rises.
- If $k$ rises, the real cost of risky money is larger, so the optimal $\lambda^*$ falls.
- If portfolio risk becomes more severe, the regulator should put more weight on the risky part of the asset payoff and require more bank equity.

**Economic message.** The capital requirement is not a mechanical hatred of leverage. It balances the funding benefit of bank liabilities against the real disruption caused by risky money.

## 3.6 Private information inside formal banking

Now suppose the bank privately observes at date 1:

- whether its loan portfolio is good or bad;
- whether its outside investment opportunity is valuable.

There are four date-1 bank types. A key result is that the optimal formal-bank contract remains the same as under symmetric information. The bank commits ex ante to issue the same claim regardless of its realized type:

$$(\mu, \lambda) = (1, \lambda^*). \tag{8}$$

**Why pooling is optimal.** A bank with a bad portfolio would like to mimic a good bank that raises cash. The regulator minimizes adverse-selection costs by forcing the bank to pool across future types. This is feasible in formal banking because the regulator can impose an ex ante target leverage before the bank observes its private information.

**Real-world interpretation.** Banks normally back deposits with large, diversified balance sheets rather than using a separate regulated subsidiary for each asset class and information state. This pooling makes formal bank deposits less information sensitive than claims traded in opaque secondary markets.

## 3.7 Shadow banking for a fixed regulatory leverage

Let the regulator impose a formal risk-adjusted leverage $\lambda$. The bank can still secretly sell some of the residual risky payoff $(1-\lambda)l$ to the MMF. Denote the shadow-banking risky pledge by $\widetilde{\lambda} \in [0, 1-\lambda]$.

The MMF is uninformed and quotes a competitive linear price $r$ per unit of risky payoff. A good bank with no valuable outside opportunity will not sell at any price below one. A bad bank mimics sellers. Therefore, conditional on trade, the seller is good only when the bank is good and also has a valuable outside opportunity. The competitive price is:

$$r = \frac{pq}{1-p+pq}. \tag{9}$$

Since $q < 1$, we have $r < p$. This is the lemons discount in the shadow market.

A good bank with a valuable outside opportunity chooses $\widetilde{\lambda}$ to maximize:

$$(p\lambda + r\widetilde{\lambda})l + L - I + f\big((p\lambda + r\widetilde{\lambda})l + L - I\big) + (1 - \lambda - \widetilde{\lambda})l. \tag{10}$$

Ignoring corner constraints, the first-order condition can be written using $\varphi = (f')^{-1}$:

$$(p\lambda + r\widetilde{\lambda})l = \varphi\left(\frac{1-p}{pq}\right) + I - L. \tag{11}$$

**Economic interpretation.** A lower formal leverage $\lambda$ raises the desired shadow pledge $\widetilde{\lambda}$. In plain language, tighter capital requirements encourage the bank to undo the constraint through the shadow market.

## 3.8 Thresholds for shadow activity

Define two thresholds:

$$\overline{\lambda} = \frac{\varphi\left(\frac{1-p}{pq}\right) + I - L}{pl}, \tag{12}$$

$$\underline{\lambda} = \frac{\varphi\left(\frac{1-p}{pq}\right) + I - L - rl}{(p-r)l}. \tag{13}$$

Then:

- If $\lambda \geq \overline{\lambda}$, the shadow-banking sector is inactive: $\widetilde{\lambda} = 0$.
- If $\lambda \leq \underline{\lambda}$, the bank sells its entire residual risky stake in the shadow sector: $\widetilde{\lambda} = 1 - \lambda$.
- For $\lambda \in (\underline{\lambda}, \overline{\lambda})$, shadow banking is partially active.

**Economic message.** Shadow banking becomes active only when the formal capital requirement is tight enough. Adverse selection lowers the shadow price $r$, which makes shadow funding costly and can therefore discipline shadow activity.

## 3.9 Why tighter requirements can be Pareto dominated

For regulatory leverages in the intermediate region $[\underline{\lambda}, \overline{\lambda}]$, the bank adjusts $\widetilde{\lambda}$ so that its date-1 cash is unchanged. A reduction in formal leverage is offset by an increase in shadow leverage. For a small decrease $d\lambda > 0$ in formal leverage, the required increase in shadow leverage is:

$$d\widetilde{\lambda} = \frac{p}{r} d\lambda > d\lambda, \tag{14}$$

because $r < p$. Thus, the bank transfers more total risk to raise the same cash.

**Economic interpretation.** The shadow sector is a worse place to refinance bank assets because the MMF requires a lemons discount. If the regulator pushes activity out of the formal sector and into the shadow sector, households can end up bearing more risk even though the formal bank looks safer.

## 3.10 Optimal regulation with shadow banking

The regulator chooses $\lambda$ while anticipating the bank's shadow-banking response.

There are three regimes:

- **Low externality regime.** If the real-side externality is small, the perfect-enforcement leverage $\lambda^*$ is high enough that shadow banking is inactive. The optimal regulation is unchanged.
- **Intermediate externality regime.** Shadow banking is a relevant threat. The regulator relaxes the capital requirement relative to the perfect-enforcement benchmark, choosing a higher leverage near $\overline{\lambda}$ so that the bank does not use the shadow sector. This sacrifices some formal safety but avoids inefficient risk transfer through shadow banking.

- **High externality regime.** There are two local optima. One is a looser formal requirement with no shadow banking. The other is a tighter formal requirement that induces full shadow banking. Either can be globally optimal depending on parameters.

**Main theoretical lesson.** Once enforcement is imperfect, the regulator may optimally loosen formal bank capital requirements. The objective is not to make banks riskier for its own sake; it is to keep risky money creation within the regulated sector, where it can be done with less adverse-selection waste.

# 4 Regulatory mechanism and model credibility

## 4.1 What makes the result work

The result rests on four linked ingredients:

1. **Money-like bank liabilities create externalities.** Risky bank liabilities reduce the usefulness of deposits and MMF shares as settlement media. This generates a welfare cost that banks do not fully internalize.
2. **Bank capital has an opportunity cost.** If banks retain more equity inside the banking sector, less date-1 cash is available for valuable nonpledgeable investments outside the bank. This creates a real cost of very tight capital requirements.
3. **Formal banking allows commitment.** A regulator can force the bank to pool across private-information states and issue a common claim. This reduces adverse-selection costs.
4. **Shadow banking is a spot market.** Once banks are privately informed, they can trade in an opaque market with MMFs. The MMF prices claims at a lemons discount, so shadow funding transfers more risk per dollar raised.

The counterproductive-regulation result is not just "banks avoid rules." It is sharper: evasion occurs through a market with worse informational properties, so the evasion itself changes the social cost of financing.

## 4.2 Why formal banking is socially preferred to shadow banking

Formal banking and shadow banking both fund risky loans with money-like liabilities. The difference is the quality of risk transfer.

Under formal banking, the regulator can choose a claim before the bank's private information arrives. This makes pooling possible. Under shadow banking, the MMF buys claims after the bank has private information. The MMF therefore suspects that the bank may be selling lemons and quotes a lower price. To raise a given amount of cash, the bank must sell a larger risky claim.

This yields the central ranking:

cash raised per unit of risk is higher in formal banking than in shadow banking.

Therefore, forcing activity from formal banking into shadow banking can reduce welfare even if formal bank leverage falls.

## 4.3 How to interpret the MMF

The MMF is a reduced-form representation of the shadow-banking chain. In practice, precrisis shadow banking involved MMFs, asset-backed commercial paper conduits, asset-backed securities, liquidity guarantees, and accounting rules. The model collapses this chain into one key function:

$$\text{MMF liabilities} \quad \longleftrightarrow \quad \text{money-like claims backed by bank-risk exposures.}$$

The institutional details are not the focus. What matters is that banks can pledge additional risk outside the regulatory perimeter and that the claims ultimately look money-like to nonfinancial agents.

## 4.4 Adverse selection: good and bad

Adverse selection in the shadow sector has two effects:

- **Disciplining effect.** The lemons discount makes shadow funding expensive, reducing the amount of shadow banking.
- **Risk-transfer effect.** Conditional on raising cash in the shadow sector, the bank must sell a larger claim, transferring more risk per dollar raised.

The paper's equilibrium result is subtle. When the regulator optimally chooses formal leverage, the disciplining effect dominates at the margin. More adverse selection makes the shadow sector less attractive, so equilibrium welfare rises under imperfect enforcement. But if regulation is inefficiently tight and shadow banking is helping undo a bad rule, adverse selection becomes costly because it makes this undoing inefficient.

## 4.5 Regulatory interpretation

The paper offers a warning against regulating only the measured balance sheet of formal banks. A rule that lowers measured bank leverage can fail if it raises unmeasured leverage through shadow vehicles. The relevant policy object is not:

$$\text{formal bank leverage alone,}$$

but rather:

$$\text{effective risk exposure of all money-like liabilities.}$$

The optimal capital requirement may therefore need to be looser than the perfect-enforcement benchmark if the alternative is larger, more opaque, and more information-sensitive shadow intermediation.

## 4.6 Limitations to keep in mind

- The model is highly stylized: one bank, one MMF, one household, one entrepreneur, and two dates.
- Shadow banking is represented as an unobserved trade between the bank and MMF, abstracting from the full legal structure of securitization, repo, guarantees, and conduits.
- The main model does not fully model runs, fire sales, deposit insurance, bailouts, or dynamic regulatory learning.
- The bank's investment opportunity and the entrepreneur's adjustment cost are reduced-form devices. They make the trade-off transparent, but they are not a full general-equilibrium model of credit allocation.
- The regulator's observability is binary: formal trades are observed, shadow trades are not. Real-world regulation may partially observe some shadow activities.

These limitations do not undermine the core point. They clarify that the paper is not an empirical estimate of optimal capital ratios; it is a theoretical warning about how imperfect enforcement can reverse the intended effect of capital regulation.

# 5 Propositions and figures

## 5.1 Proposition 1: optimal capital regulation with perfect enforcement

**Read:**

- The regulator fully allows pledging of the safe payoff: $\mu^* = 1$.
- The regulator restricts pledging of the risky payoff to $\lambda^* \in (0, 1)$.
- $\lambda^*$ solves the marginal condition:

$$qf'(\lambda^* pl + L - I) = \lambda^* k(1-p)l.$$

**Economic message.** Capital regulation is a Pigouvian correction for the externality created by risky money-like liabilities. The safe part of the asset can be pledged freely; the risky part must be partly retained by bank equity.

## 5.2 Figure 1: bank balance sheet after initial cash transfers

**Read:**

- The asset side is the bank's loan portfolio.
- The liability side contains a household deposit claim and a residual equity claim.

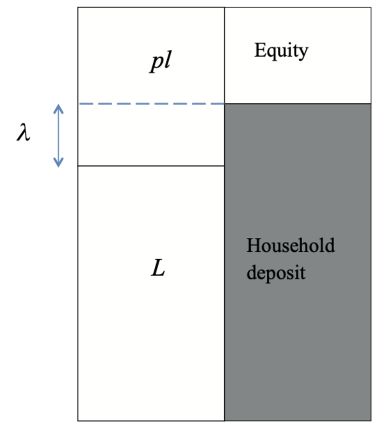


**Figure 1**
Bank balance sheet after initial cash transfers are made

- The household claim is senior and includes a safe tranche backed by $L$ plus a risky tranche backed by $\lambda^* l$.

**Economic message.** The capital requirement is not a cap on all debt in the same way. It is a cap on the risky part of the portfolio that can back money-like liabilities. The model maps naturally into a risk-adjusted leverage requirement.

## 5.3 Proposition 2: private information does not change formal-bank regulation

**Read:**

- Even when the bank privately observes portfolio quality and outside-investment quality, the optimal formal-bank security is still $(1, \lambda^*)$.
- The bank must commit to the same leverage regardless of its realized date-1 type.

**Economic message.** Formal regulation is valuable partly because it creates commitment and pooling. Pooling is a way to avoid the adverse-selection problem that would arise if each bank type chose its own financing ex post.

## 5.4 Lemma 4: threshold structure of shadow banking

**Read:**

- If the formal regulatory leverage is high enough, $\lambda \geq \overline{\lambda}$, the bank does not use shadow banking.
- If the formal regulatory leverage is low enough, $\lambda \leq \underline{\lambda}$, the bank sells all residual risky cash flow in the shadow sector.
- In between, shadow banking partially offsets the formal capital requirement.

**Economic message.** Shadow banking is an endogenous response to tight regulation. It is not an exogenous sector with fixed size.

## 5.5 Lemma 5: tighter regulation can be Pareto dominated

**Read:**

- In the intermediate region, the bank undoes a reduction in formal leverage by increasing shadow leverage.
- Because $r < p$, a dollar of shadow funding requires more risk transfer than a dollar of formal bank funding.
- A looser requirement can leave the bank no worse off while reducing household risk.

**Economic message.** A stricter formal rule may improve the regulated balance sheet while worsening the total risk borne by holders of money-like claims.

## 5.6 Proposition 3: optimal capital regulation with shadow banking

| Regime | Regulatory choice | Economic meaning |
|---|---|---|
| Low externality | Choose the perfect-enforcement leverage $\lambda^*$; shadow banking remains inactive. | The capital requirement is not tight enough to induce costly regulatory arbitrage. |
| Intermediate externality | Relax the capital requirement relative to $\lambda^*$ to deter shadow banking. | The regulator intentionally keeps activity in formal banking because formal funding transfers less risk per dollar raised. |
| High externality | Two local optima: a loose/no-shadow solution and a tight/full-shadow solution. | The regulator may either keep activity visible or accept a large shadow sector; the global optimum depends on parameter values. |

**Economic message.** Once shadow banking is possible, optimal regulation may be nonmonotone. The best response to shadow banking need not be tougher formal capital requirements.

## 5.7 Figure 2: surplus under the high-externality case

**Read:**

- The figure illustrates the case with two local maxima.
- One maximum occurs at a relatively loose capital requirement that shuts down shadow banking.
- The other occurs at a tight capital requirement that induces full shadow banking.
- The middle range is unattractive because the bank partially offsets regulation in an inefficient way.

**Economic message.** The welfare objective can have multiple peaks. This creates a policy trade-off between visible leverage inside banks and hidden leverage outside banks.

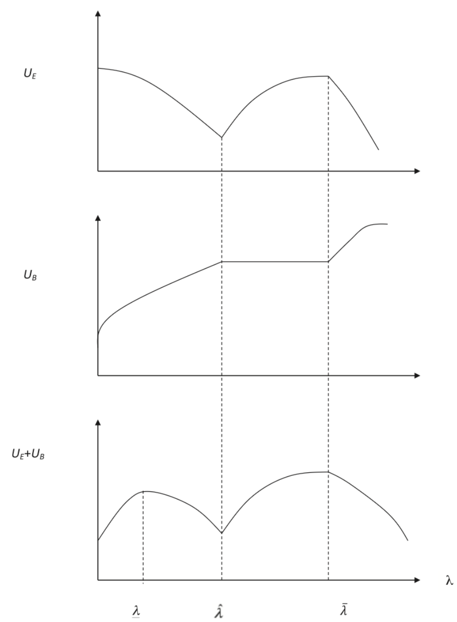


**Figure 2**
Surplus in Proposition 3 case c

## 5.8 Proposition 4: adverse selection can improve welfare under optimal regulation

**Read:**

- A compensated reduction in $p$ or $q$ increases adverse selection in the shadow sector.
- Under perfect enforcement, these compensated changes leave welfare unchanged by construction.
- Under imperfect enforcement, they raise total surplus because they reduce shadow-banking activity.

**Economic message.** If regulation is chosen optimally, some opacity in shadow markets can be socially useful because it makes regulatory arbitrage less attractive. This is not a general defense of opacity; it is a specific equilibrium result under optimal regulation.

## 5.9 Proposition 5: inefficient regulation can make shadow banking desirable

**Read:**

- Suppose the implemented capital requirement is inefficiently tight.
- If adverse selection is small, shadow banking can improve welfare by helping banks bypass an excessive constraint.
- If adverse selection is large, shadow banking is costly because it bypasses the constraint with excessive risk transfer.

**Economic message.** Whether shadow banking is good or bad depends partly on whether the underlying regulation is efficient. If the regulation is too tight, some arbitrage can be welfare-improving; if the regulation is broadly efficient, arbitrage is costly.

### 5.10 Extensions: transaction costs and asset encumbrance

**Alternative illiquidity.** The paper shows that the main logic does not require literal adverse selection. Convex transaction costs in the shadow sector can produce similar effects if marginal shadow-trading costs do not rise too quickly relative to the decline in marginal investment returns.

**Asset encumbrance.** The paper also sketches an extension in which the regulator cannot distinguish safer and riskier assets. Banks may then pledge their better collateral in the shadow sector and leave lower-quality claims in the formal balance sheet. This connects the model to concerns about excessive asset encumbrance and "runs on collateral."

## 6 Short summary

- The paper develops a theory of bank capital regulation when bank liabilities serve as money-like claims.
- Risky bank liabilities create a real externality because they reduce the reliability of the household's purchasing power and disrupt trade with the entrepreneur.
- An unregulated bank chooses maximum leverage because it values date-1 cash but does not internalize the full real-side cost of risky money.
- With perfect enforcement, the regulator imposes a binding risk-adjusted leverage cap $\lambda^*$.
- With private information but formal regulation, the same capital requirement remains optimal because the bank can commit to pooling across future types.
- Shadow banking appears when the bank can secretly sell residual risky cash flows to an MMF that also issues money-like liabilities.
- The shadow sector is a spot market with adverse selection, so the MMF pays a discounted price $r < p$.
- Tighter formal requirements can induce more shadow banking, and shadow funding transfers more risk per dollar raised.
- Therefore, the optimal regulator may relax capital requirements to keep activity in the regulated banking sector.
- Adverse selection in shadow banking is disciplining when regulation is optimal, but costly when regulation is inefficiently tight.

## 7 Conclusion

## 7.1 Core conclusion

Plantin's central conclusion is that bank capital regulation must be designed with regulatory arbitrage in mind. In a world with perfect enforcement, capital requirements reduce the externality from risky money-like bank liabilities. In a world with shadow banking, the same tightening can push risk transfer into opaque markets and increase the total risk borne by holders of money-like claims.

## 7.2 Economic intuition

The key intuition is that formal banks and shadow banks are not just substitutes in quantity; they differ in informational quality. Formal bank regulation can impose ex ante pooling and reduce adverse selection. Shadow banking takes place after banks are informed, so buyers demand a lemons discount. When a capital rule pushes financing into this lower-quality market, the bank must sell more risk to raise the same cash.

## 7.3 Takeaway

The paper reverses a policy instinct. More stringent formal bank capital requirements are not always safer. If the regulatory perimeter is porous, a tighter formal rule may expand shadow banking and raise effective risk-adjusted leverage in the financial system. The relevant policy target is the safety of all money-like liabilities, not only the measured leverage of regulated banks.